The following theorem says that characters of irreducible representations of G form an orthonormal basis of $F_c(G, \mathbb{C})$ under this inner product.

Theorem 4.5.1. *For any representations V, W*

$$(\chi_V, \chi_W) = \dim \mathrm{Hom}_G(W, V),$$

and

$$(\chi_V, \chi_W) = \begin{cases} 1, & \text{if } V \cong W, \\ 0, & \text{if } V \ncong W \end{cases}$$

if V, W are irreducible.

Proof. By the definition

$$(\chi_V, \chi_W) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)} = \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \chi_{W^*}(g)$$
$$= \frac{1}{|G|} \sum_{g \in G} \chi_{V \otimes W^*}(g) = \mathrm{Tr}\,|_{V \otimes W^*}(P),$$

where $P = \frac{1}{|G|} \sum_{g \in G} g \in Z(\mathbb{C}[G])$. (Here $Z(\mathbb{C}[G])$ denotes the center of $\mathbb{C}[G]$.) If X is an irreducible representation of G, then

$$P|_X = \begin{cases} \mathrm{Id} \text{ if } X = \mathbb{C}, \\ 0, \quad X \neq \mathbb{C}. \end{cases}$$

Therefore, for any representation X the operator $P|_X$ is the G-invariant projector onto the subspace X^G of G-invariants in X. Thus,

$$\mathrm{Tr}\,|_{V \otimes W^*}(P) = \dim \mathrm{Hom}_G(\mathbb{C}, V \otimes W^*)$$
$$= \dim (V \otimes W^*)^G = \dim \mathrm{Hom}_G(W, V).$$

□

Theorem 4.5.1 gives a powerful method of checking if a given complex representation V of a finite group G is irreducible. Indeed, it implies that V is irreducible if and only if $(\chi_V, \chi_V) = 1$.

Problem 4.5.2. Let G be a finite group. Let V_i be the irreducible complex representations of G.